METRIC

MIL-STD-1399
SECTION 300, PART 3
9 April 2026

DEPARTMENT OF DEFENSE
INTERFACE STANDARD
SECTION 300, PART 3
LOW VOLTAGE ELECTRIC POWER DIRECT CURRENT

AMSC N/A FSC 1905

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This part defines the standard interface requirements and verification methods for the design of shipboard power sources and user equipment that will utilize shipboard direct current (DC) electric power at approximately 850 volts. Power sources are power conversion equipment within the electric power system. User equipment are loads and power conversion equipment within the electric power system.

3. In any system involving a power source, distribution network, and load (user equipment), the characteristics at the system and user equipment interface are mutually dependent on the design and operation of both. In order for the electric power system to perform within the established tolerances, it is necessary to place constraints on the power source, the distribution system, and the user equipment. This interface standard defines the electric power system characteristics for shipboard DC interfaces. User equipment constraints are also established.

4. The technical content first delineates the characteristics of the shipboard DC electric power system at the interface in terms of voltage, continuity, and voltage distortion. Constraints on user equipment design, which are necessary to achieve shipboard compatibility with and to assure these characteristics, are then established. Test requirements are specified to verify conformance of user equipment to this standard. Power source, including energy storage, requirements and verification methods are also specified.

5. Naval Sea Systems Command (NAVSEA) will consider the mission requirement of the user equipment being developed or acquisitioned in determining if the equipment is mission critical equipment (MCE).

6. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, D.C. 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This military standard section establishes electrical interface characteristics for shipboard equipment utilizing or providing low voltage direct current (LVDC) electric power at 850 (nominal) volts for type I and type III power and 875 (nominal) volts for type II power (see 4.2 and 6.2) to ensure compatibility between electrical power sources and user equipment. Characteristics of the electrical interfaces as well as interface requirements and interface verification methods for electrical power sources and user equipment are provided. For equipment with multiple electrical power interfaces, this standard only applies to those interfaces utilizing the applicable voltages generated by the shipboard electrical distribution systems.

1.2 Classification. This standard defines type I, type II, and type III power quality requirements for nominal line-to-line voltages of shipboard direct current (DC) electric power as listed in table I. Additional types and supply voltages may be defined in the future.

Type I - Type I power (example figure 1) is derived by rectification of alternating current (AC) power as specified in MIL-STD-1399-300-1 or MIL-STD-1399-300-2. Rectification can be by simple diode or silicon-controlled rectifier (SCR) systems or more sophisticated active-front-end (AFE) rectifiers. The type I ship service DC power bus does not have energy storage so the ability to power dynamic loads is dictated by the limitation of the MIL-STD-1399-300-1 or MIL-STD-1399-300-2 interface to the AC-DC converter providing the type I power. The interface of the AC-DC rectifier to the AC system must meet the interface requirements of MIL-STD-1399-300-1 or MIL-STD-1399-300-2 as appropriate.

Type II - Type II power (example figure 2) systems have energy storage, typically batteries, either directly across the ship service DC power bus or connected through a power converter that does not tightly regulate the bus voltage. The characteristics of the voltage at the type II user equipment interface are primarily defined by the terminal voltage of the energy storage system and, as such, have a wider voltage range than type I and type III. The AC-DC rectifier providing power to the type II bus serves to charge the energy storage and additionally may provide power directly to user equipment. The interface of the AC-DC rectifier to the AC system must meet the interface requirements of MIL-STD-1399-300-1 or MIL-STD-1399-300-2 as appropriate.

Type III - Type III power (example figure 3) is derived from active conversion of ship system AC power in conjunction with active conversion of DC voltage on energy storage devices. The distinction between type II and type III power is that type II voltage is primarily determined by the voltage of the energy storage element, while type III power has a power converter between the energy storage element and the bus that provides tightly regulated voltage. Type III power is intended to provide more dynamic capability through the control of the flow of power from energy storage via DC/DC converter-based control. This may be supplemented by employment of an actively controlled rectifier interface to the AC primary distribution to control the flow of power from the AC bus. The interface of the AC-DC rectifier to the AC system must meet the interface requirements of MIL-STD-1399-300-1 or MIL-STD-1399-300-2 as appropriate.

1.2.1 Non-standard voltages. If user equipment requires voltage magnitudes or qualities different than defined at these interfaces, these user equipment will need to provide this via internal power conversion, which will be part of the mission system. In such cases, the requirements of this document or MIL-STD-1399-300-1 or MIL-STD-1399-300-2, as applicable, must be met at the interface between the mission load system and the ship AC or DC voltage interface.

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TABLE I. Nominal DC system voltages.

Nominal System Voltage	Type	Description
850 VDC	Type I	Direct rectification of AC voltage without energy storage.
875 VDC	Type II	Energy storage bus. Direct connection to an energy storage bus or connected through a power converter that does not tightly regulate the bus voltage. Voltage varies in accordance with the terminal voltage of the energy storage media.
850 VDC	Type III	Controlled rectifier and controlled converter interfacing to energy storage on the bus.

1.3 Electrical interface. The basic characteristic and constraint categories concerned with this interface are shown symbolically on figures 1 through 3. The source interface is located at the output terminals of the electric power source(s). The load interface is located at the input terminals of the electrical load(s).

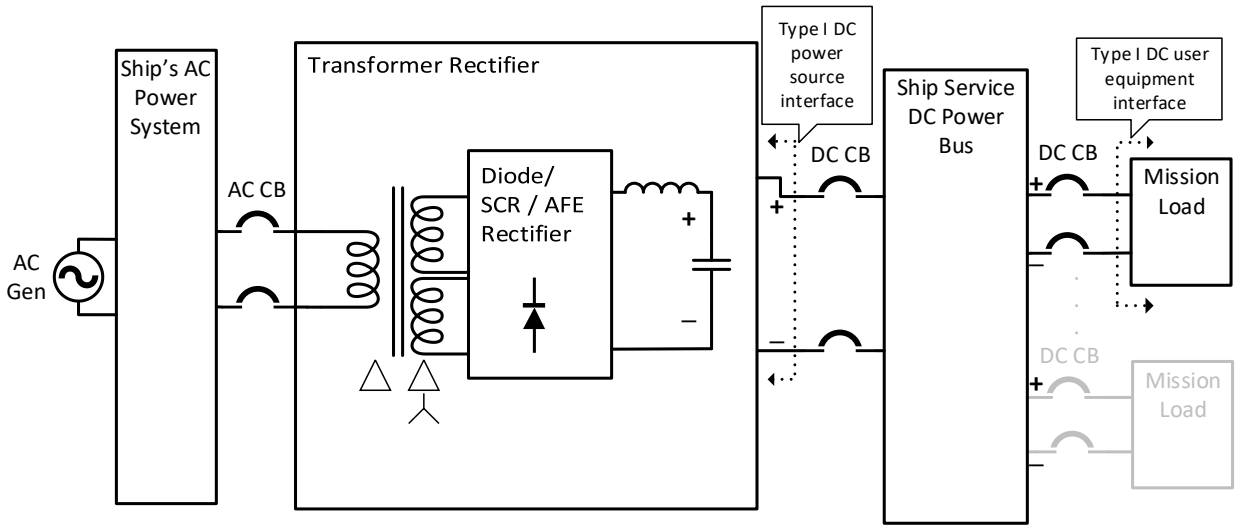


FIGURE 1. Example of type I DC interfaces – diode rectifier, SCR, or AFE rectifier (typical, topologies will vary).

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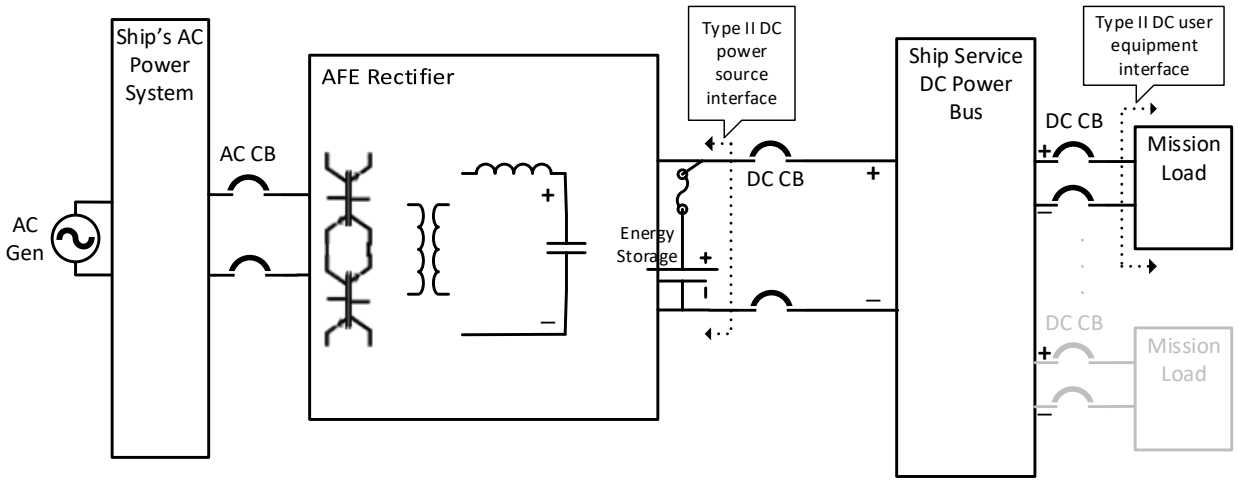


FIGURE 2. Example of type II DC interfaces – energy storage bus (typical, topologies will vary).

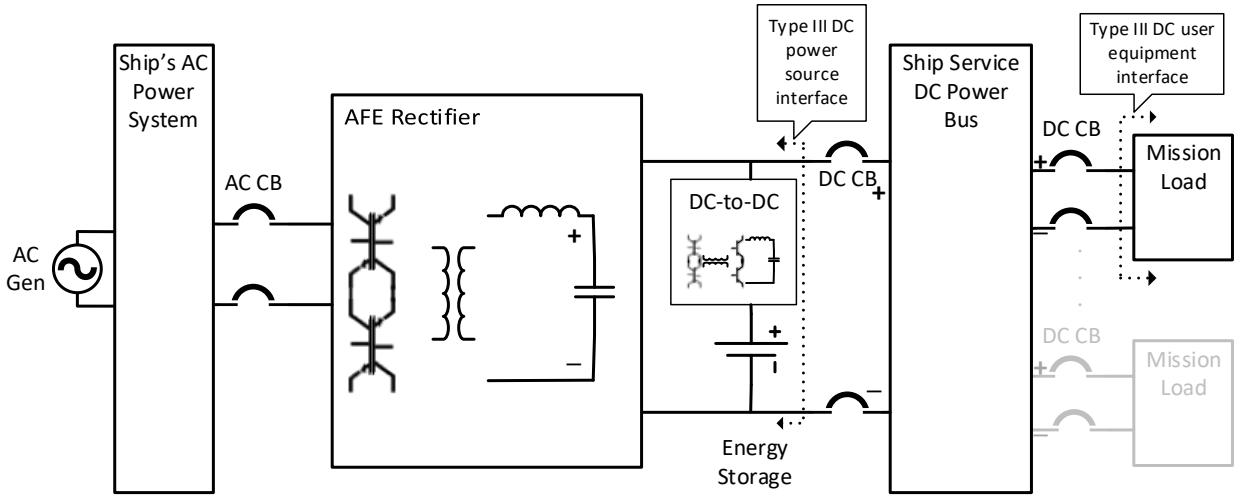


FIGURE 3. Example of type III power interfaces – controllable interfaces to AC bus and energy storage (typical, topologies will vary).

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include publications cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

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MIL-STD-1310 - Shipboard Bonding, Grounding, and other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety

MIL-STD-1399 - Interface Standard for Shipboard Systems

MIL-STD-1399-300-1 - Low Voltage Electric Power, Alternating Current

MIL-STD-1399-300-2 - Medium Voltage Electric Power, Alternating Current

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-2036 - Preparation of Electronic Equipment Specifications

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9300-AF-PRO-020 - Electrical Systems Design Criteria and Practices (Surface Ships) for Preliminary and Contract Design

(Copies of this publication are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). This publication is available for ordering (hard copy) via the Naval Logistics Library [NLL] at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

IEEE

A. Riccobono and E. Santi, “Comprehensive Review of Stability Criteria for DC Power Distribution Systems,” IEEE Transactions on Industry Applications, Vol. 50, No. 5, pp. 3525-3535, Sept.-Oct. 2014, DOI: 10.1109/TIA.2014.2309800

(Copies of this publication are available online at https://ieeexplore.ieee.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61000-4-5 - Electromagnetic Compatibility (EMC) - Part 4-5: Testing and Measurement Techniques - Surge Immunity Test

(Copies of this publication are available online at www.iec.ch.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Common mode current. The sum of the instantaneous currents of the positive and negative conductors out of the interface divided by two. The common mode current is typically depicted as a function of frequency. Common mode currents have return paths outside the set of conductors, typically through the ship’s hull or cable shields.

3.2 Common mode impedance. The common mode impedance at the interface, referenced to the ground plane, is the common mode voltage at the interface divided by the common mode current at the interface. The common mode impedance is typically depicted as a function of frequency.

3.3 Common mode voltage. The average of the two voltages from the positive rail to the ground plane and the negative rail to the ground plane at the interface of interest,

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$V_{cmm} = \frac{V_{PositiveRail}+V_{NegativeRail}}{2}$ so that $V_{NegativeRail} = \frac{-V_{Nominal}}{2} + V_{cmm}$ and $V_{PositiveRail} = \frac{V_{Nominal}}{2} + V_{cmm}$. The common mode voltage is typically depicted as a function of frequency.

3.4 Control negotiations. Control interactions between user equipment or power sources and the ship's power system using a standard protocol that result in the establishment of limits for the behavior of the user equipment or power sources. Such control negotiations may apply to large loads and large pulsed loads where the power system may require reconfiguration before the power needs of the user equipment can be met.

3.5 DC user equipment interface. The user interface regarding the characteristics of DC power is located at the input power terminals of user equipment. Figures 1 through 3 depict the interface between the electric power system and user equipment. This interface is the location wherein the electric power system characteristics and the user equipment requirements apply.

3.6 Electric power system. The electric power generation and distribution system including generation, cables, transformers, switchboards, switches, protective devices, power electronic converters, and energy storage devices up to the user equipment interface.

3.7 Electric power system ground. A plane or surface used by the electric power system as a common reference to establish zero potential. Usually, this surface is the metallic hull of the ship. Sometimes, simply referred to as ground.

3.7.1 Grounding (of electric systems/equipment). The process of establishing a minimum impedance current path between a selected electric circuit or equipment and ground potential.

3.7.2 Ungrounded electric power system. A system that is not intentionally connected to the metal structure or the grounding system of the ship, except for test purposes. An ungrounded electric power system can continue to perform normally if one line conductor becomes solidly grounded.

3.8 Emergency conditions. Unexpected occurrences of a serious nature that may result in electrical power system deviations. Emergency conditions include, but are not limited to, battle damage and malfunction or failure of equipment. Conditions may include power interruptions, voltage excursions, and/or loss of energy storage. Emergency conditions characteristics are provided in table II and table III.

3.9 Function normally. Operate in conformance with equipment and system specifications and with no increase in safety risk.

3.10 Low voltage, direct current. DC voltages at or below 1,000 volts.

3.11 Mission critical equipment. Equipment that is part of mission critical systems and required to operate through emergency conditions (see T9300-AF-PRO-020).

3.12 No-break supply. A source of power to a user equipment designed not to have power interruptions. A no-break supply can be implemented via methods such as auctioneering diodes or highly reliable dedicated power conversion with two sources of power, one of which may be energy storage.

3.13 Nominal system voltage. The design system DC voltage measured between the positive and negative conductors (line-line) as specified in table II and table III. Nominal system voltage is used as a reference for establishing other power quality requirements.

3.14 Normal operating mode. A particular condition of a piece of equipment where the equipment is fulfilling one of its defined energized operational functions. The deenergized or off state is not a normal operating mode in this context.

3.15 Phase slope method. A method of assessing the damping of a system's electrical behavior from a stability perspective. Phase slope is a simple rise-over-run calculation on a logarithmic frequency scale, such that:

$$\text{Phase Slope} = \left|\frac{phase_2 - phase_1}{\log_{10} f_2 - \log_{10} f_1}\right|$$

where $phase_{1,2}$ are the measured phase values of an impedance at frequencies $f_{1,2}$.

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3.16 Power interruption. A decrease in power system voltage to below a level that is treated as a complete loss of input power by user equipment. Power interruption is typically caused by failures such as the loss of a power source, a bus fault, or severe overload conditions. Power interruptions are characterized as an emergency condition.

3.17 Power source. A provider of power to the DC distribution equipment under normal operation.

3.18 Power source impedance. The transfer function of power source output voltage negated relative to output current (e.g., -V(s)/I(s)). Power source impedance is represented in terms of gain and phase angle of the frequency response associated with this transfer function. Power source impedance gain is equal to voltage amplitude divided by current amplitude as a function of frequency (f) and is represented in ohms. Power source impedance phase angle is equal to negated voltage phase angle minus current phase angle as a function of frequency.

3.19 Power source maximum root mean square (rms) current ripple. The non-DC component of the current waveform of the power source expressed at individual frequencies. This can be expressed in amperes or in decibels micro-amperes (dBμA).

3.20 Power source voltage variations.

3.20.1 Power source maximum steady-state voltage ripple (peak-to-peak percent of nominal system voltage). The maximum peak-to-peak value of the non-DC component of the voltage measured at the output of an online power source.

3.20.2 Power source transient voltage tolerance. The maximum departure from nominal voltage due to transient loading and unloading of user equipment and transients due to AC input disturbances that feed through the power source.

3.20.3 Power source normal service voltage range (0 to rated power). The allowable variation in the system DC voltage measured at the output of an online power source in normal droop operation mode, normal voltage regulation mode, and constant power mode. This range does not account for potential voltage loss in the distribution system.

3.20.4 Power source normal service voltage range (charged to discharged). The allowable variation in the system DC voltage measured at the output of a type II power source during the operation across the state of charge of the battery. This range does not account for potential voltage loss in the distribution system but does include internal voltage drops due to load current.

3.21 Pulsed load. User equipment that draws current in a short duration pulse, characterized by rapid rates of change during the positive and negative edges of the current pulse. The pulses drawn by the user equipment may be regular and repetitive or infrequent. Attributes for infrequent pulsed loads include user equipment surge/inrush current and user equipment ramp load. Attributes of repetitive pulsed load are pulsed power magnitude/frequency limits (frequency domain), power total signal distortion (TSD) limit (frequency domain), and pulsed power deviation limit (time domain).

3.22 Rated current. The rated maximum power of a user equipment or power source divided by the nominal system voltage.

3.23 Ship service DC power bus. A DC power distribution system providing DC voltage of type I, type II, or type III to user equipment. This bus is part of the ship's overall power distribution system.

3.24 Symmetric auctioneering diodes. A configuration for providing uninterruptible transfer of power between two sources of power to user equipment via diodes on both the positive and negative conductors. This configuration helps control common mode currents, but in certain fault conditions, specifically a double ground fault, may result in user equipment being subjected to a voltage equal to the sum of the two source voltages unless compensated with a reverse bias diode in parallel with the forward bias diode on the negative rail. No-break sources using antisymmetric auctioneering diodes, where the diodes are only in one rail, do not exhibit the same overvoltage conditions but can result in excessive common mode currents.

3.25 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

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3.26 Temperature stability. Steady-state temperature stability is achieved when the variation between successive temperature measurements at the same location does not change by more than 1 °C after 30 minutes.

3.27 User equipment. Any system or component installed in the ship that uses power supplied directly by the DC power system.

3.28 User equipment input admittance. The transfer function of user equipment input current relative to input voltage (e.g., I(s)/V(s)). User equipment input admittance is represented in terms of gain and phase angle of the frequency response associated with this transfer function. User equipment input admittance gain is equal to current amplitude divided by voltage amplitude as a function of frequency (f) and is represented in mhos or siemens. User equipment input admittance phase angle is equal to current phase angle minus voltage phase angle as a function of frequency.

3.29 User equipment input impedance. The transfer function of user equipment input voltage to input current (e.g., V(s)/I(s)). User equipment input impedance is the inverse of user equipment input admittance.

3.30 User equipment current and power variations.

3.30.1 User equipment current ripple frequency content. The non-DC component of the current waveform of user equipment expressed at individual frequencies. This can be expressed in amperes or in dBμA. Depending upon the load behavior for each operational state, this waveform may change with time as well as frequency content.

3.30.2 User equipment maximum rms current ripple. The maximum rms magnitude of the steady-state, non-DC component defined as $\sqrt{I_{rms(1)}^2 + I_{rms(2)}^2 + I_{rms(3)}^2 + I_{rms(n)}^2}$, where $I_{rms(n)}$ represents the rms current at each of the individual frequencies that appear in the user equipment's current waveform. This is also referred to as current TSD.

3.30.3 User equipment maximum inrush current. A sudden change in line current to user equipment that occurs during start-up or after a power interruption or as a result of a change to the operating mode. Typically, the surge current will rise to a maximum value in a few milliseconds and decay to rated value in several milliseconds to several seconds. The limit is evaluated as the ratio of the highest peak surge/inrush current to the rated current of the equipment.

3.31 User equipment voltage variations.

3.31.1 User equipment maximum steady-state voltage ripple (peak-to-peak percent of nominal system voltage). The maximum peak-to-peak value of the non-DC component of the voltage measured at the input of a user equipment.

3.31.2 User equipment normal service steady-state voltage range. The range of steady-state DC voltage that, when applied to user equipment (including voltage ripple), the user equipment is expected to function normally. This range accounts for possible voltage drop in the distribution system as well as the power source normal service voltage range. For type II power, this is characterized as user equipment normal service voltage range, overcharged to discharged. This is the degree to which voltage can vary solely due to varying states of charge of the energy storage system.

3.31.3 User equipment voltage spike. A voltage change or impulse of very short duration (less than or equal to 1 millisecond) that exceeds the user equipment voltage transient tolerance. For test purposes of this standard, in the voltage spike test of 5.3.5, the voltage increases to maximum voltage in no more than 1.2 microseconds and decays to ½ the peak voltage applied in approximately 50 microseconds. The voltage spike may be positive or negative and occur either from line-to-line or from line-to-ground. Voltage spikes in shipboard power systems are generally of an oscillatory nature and not unidirectional like those often used in testing.

3.31.4 User equipment voltage transient recovery time. The time elapsed from the instant the voltage first goes outside the user equipment normal service steady-state voltage range until the instant when the voltage recovers and remains within the user equipment normal service steady-state voltage range.

3.31.5 User equipment voltage transient tolerance. The allowed variation from nominal voltage expressed as a percent of the nominal voltage during transient conditions; the permitted voltage transient tolerance is provided in table II and table III.

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3.31.6 Voltage transients. A sudden change in voltage (longer than 1 millisecond) that exceeds the normal service steady-state voltage range, positively or negatively and returns to and remains within these limits within a specified recovery time, as specified in table II and table III, after the initiation of the disturbance such as large user equipment current changes.

3.32 Voltage power quality. Adherence to the voltage requirements of table II or table III, as applicable.

4. GENERAL REQUIREMENTS

4.1 Interface requirements. The specific interface requirements and constraints established herein are mandatory and shall be adhered to regarding any aspect of shipboard electrical power systems or user equipment designs to which these requirements and constraints apply, including systems and equipment design, production, and installation. This section and the basic standard are to be viewed as an integral single document for use in the design and testing of electric power systems and user equipment. MIL-HDBK-2036 may be used as guidance for tailoring requirements.

4.1.1 User equipment. User equipment shall adhere to the interfaces specified in section 5.2. User equipment shall be verified as specified (see 6.2) to adhere to the interface requirements specified through the verification methods in section 5.3. For user equipment that have multiple applicable interfaces, the requirements in section 5.2 shall apply to each interface individually.

4.1.2 Power sources. Power sources shall adhere to the interfaces specified in section 5.4. Power sources shall be verified as specified (see 6.2) to adhere to the interface requirements specified through the verification methods in section 5.5. For power sources that have multiple applicable interfaces, the requirements in section 5.4 shall apply to each interface individually.

4.1.3 Equipment acting as both power sources and user equipment. Some equipment, such as energy storage modules, may have power interfaces that sometimes act as power sources and at other times act as user equipment. When acting as a power source, the power source interface requirements shall apply (see 4.1.2). When acting as a user equipment, the user equipment interface requirements shall apply (see 4.1.1).

4.2 Nominal system voltage. Unless otherwise specified (see 6.2), the nominal system voltage shall be 850 VDC for type I and III power and 875 VDC for type II power.

4.3 Normal operating mode. The normal operating modes shall be as specified (see 6.2).

4.4 Distribution equipment. The impedance, switching, transient surge suppression, and fault clearing capability of the distribution equipment shall be designed to ensure that if power sources and user equipment adhere to this standard, then the current characteristics at the power source interface will also adhere to this standard. Similarly, if the power sources and user equipment adhere to this standard, then the voltage characteristics at the user equipment interface will also adhere to this standard.

4.5 Measurements. Unless otherwise specified (see 6.2), the following measurement methods shall be used for the tests in 5.3 and 5.5 to evaluate compliance of power, voltage, and current properties with this standard.

4.5.1 Measurement sampling rates. For all measurements, except for voltage spike, current waveform, common mode currents, and power ripple-related measurements, the sampling rate shall be constant and equal to or greater than 10 times the switching frequency of the source/user equipment under test (EUT) or 100 kilohertz, whichever is greater.

For power ripple-related tests; user equipment pulsed power deviation limit (time domain) as specified in 5.2.4.3, user equipment pulsed power magnitude/ frequency limits (frequency domain) as specified in 5.2.4.4, and user equipment power TSD limit (frequency domain) as specified in 5.2.4.5; sample rate should be high enough to accurately resolve 2 kilohertz components, including effects of the measurement device anti-alias filters. It is recommended to keep the sampling frequency below 50 kilohertz to keep the memory usage of the computer reasonable during the Discrete Fourier Transform (DFT) calculation used to assess these parameters.

The sampling rate for the voltage spike shall be 30 megahertz (MHz) or greater.

The measurement method for the current waveform test shall be as specified in MIL-STD-461, CE101. The sampling rate for common mode currents and voltages shall be as specified (see 6.2).

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4.5.2 Measurement resolution. For all measurements, the accuracy shall be ±0.5 percent for voltage and current and ±0.5 °C for temperature.

4.6 Deviations. The interfaces in this standard are based on a DC electric power system utilizing power electronic sources. To meet the intent of this section for other electric power systems, deviation of requirements will be considered. The deviation provisions specified in MIL-STD-1399 shall be adhered to during the early development stage of equipment as specified (see 6.2 and 6.3). For large loads and loads known to be challenged by specific requirements, the tailoring of requirements may be necessary. Tailoring is done with analysis of the intended system and tradeoffs between the requirements and the equipment designs needed to ensure compliance. The recommended tailoring shall be submitted for approval by NAVSEA (see 6.2 and 6.3). Equipment requiring a deviation shall have an approved deviation. Requests for deviation shall be submitted for approval by the Technical Authority (see 3.25).

4.7 Conformance tests. Test requirements to ensure conformance of user equipment to the interface characteristics and constraints specified in this standard shall be as specified (see 6.2).

5. DETAILED REQUIREMENTS

5.1 Common requirements for both user equipment and power sources.

5.1.1 Grounding.

a. System grounding. Under normal operation, the power system line-to-ground voltages will be roughly centered around ground potential. For example, an 850-volt power system will typically have ±425 volts with respect to ground under normal operation but may experience offsets of this voltage during line-to-ground faults. The DC power system will generally be ungrounded but may include grounding networks to limit common mode current and establish a high resistance path between current carrying components and ground to limit fault currents.

b. Maximum line-to-ground voltage. The maximum line-to-ground magnitude occurs when one of the conductors is faulted to ground. The unfaulted conductor's maximum voltage magnitude (not including voltage spikes or voltage ripple) with respect to ground equals the nominal system voltage plus the upper limit of the user equipment normal service steady-state voltage range (for example, for type I power, this would result in 850 volts + 0.08*850 volts = 918 VDC). For a ground fault on the negative conductor, the maximum line-to-ground voltage on the positive conductor will be a positive voltage. For a ground fault on the positive conductor, the maximum line-to-ground voltage on the negative conductor will be a negative voltage. User equipment and power sources shall remain undamaged, shall not suffer any data loss, shall not require a restart or reboot, and shall remain fully operational before, during, and after being subjected to single line-to-ground faults for any length of time. See 5.5.2 for the verification method for user equipment and power sources.

c. Equipment grounding. Power sources and user equipment shall implement equipment grounding in accordance with MIL-STD-1310 to contribute to electromagnetic compatibility and safety. Equipment grounding establishes a low impedance connection between non-current carrying metallic parts and the ground potential.

5.1.2 AC power system feeding DC power system. For type I systems, many user equipment and power source requirements are based upon the characteristics of the AC power system supplying power to the interface rectifier. The AC system source capacity, whether the interface to the rectifier is a MIL-STD-1399-300-1 interface or a MIL-STD-1399-300-2 interface, and other relevant characteristics shall be as specified for type I systems (see 6.2).

5.2 User equipment requirements. User equipment shall meet the requirements specified in 5.2.1 through 5.2.3 to ensure compatibility with the electrical power system characteristics specified in in table II and table III.

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TABLE II. Characteristic of shipboard LVDC electric power systems – type I and III power.

Characteristic	Value: Type I (interface without energy storage)	Value: Type III (interface with regulated energy storage)	Verification Method
Voltage			
Nominal system voltage	850 VDC	850 VDC	N/A
User equipment normal service steady-state voltage range	773 to 918 VDC (-9 to +8%)	790 to 884 VDC (-7 to +4%)	5.3.1
User equipment voltage transient tolerance (includes steady-state range plus transient variations)	646 to 1046 VDC (-24 to + 23%)	714 to 926 VDC (-16 to +9%)	5.3.2
User equipment voltage transient recovery time	2 seconds	0.25 second	5.3.2
User equipment maximum steady-state voltage ripple (peak-to-peak percent of nominal system voltage) 1/, 2/	14% up to 20 kHz then rolling off as shown on figure 4	14% up to 20 kHz then rolling off as shown on figure 4	5.3.1
User equipment voltage spike (spikes may be line-to-line or line-to-ground)	2500 V total including system voltage	2500 V total including system voltage	5.3.5
Emergency Conditions			
User equipment voltage positive excursion	+38% (220% for systems with symmetric auctioneering diodes)	+16% (220% for systems with symmetric auctioneering diodes)	5.3.2
User equipment voltage positive excursion duration	2 minutes	2 minutes	5.3.2
User equipment interruption voltage	-100%	-100%	5.3.4
User equipment power interruption duration, t_s	2 minutes Default value, see 5.2.2.1	2 minutes Default value, see 5.2.2.1	5.3.4
User equipment power interruption duration, t_r	0 millisecond (ms) (no-break supply) 150 ms (not on no-break supply) Default value, see 5.2.2.1	0 ms (no-break supply) 150 ms (not on no-break supply) Default value, see 5.2.2.1	5.3.4

NOTES:

1/ Unless otherwise specified (see 6.2), the individual narrowband components of the voltage ripple shall be limited on a frequency-specific basis as specified on figure 4.

2/ User equipment maximum steady-state voltage ripple is due to the ripple created by the source combined with ripple caused by pulsed loads on the power system.

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TABLE III. Characteristic of shipboard LVDC electric power systems – type II power.

Characteristic	Value: Type II (battery-based)	Verification Method
Voltage		
Nominal user equipment voltage	875 VDC Open-circuit unloaded	N/A
User equipment normal service voltage range, overcharged to discharged	680 to 1000 VDC (-22 to +14%)	5.3.1
User equipment voltage transient tolerance (includes steady state range plus transient variations)	650 to 1060 VDC (-25 to +21%)	5.3.2
User equipment voltage transient recovery time	2 seconds at 650 VDC 30 seconds at 1060 VDC	5.3.2
User equipment maximum voltage ripple (peak-to-peak percentage of nominal system voltage) 1/, 2/	14% up to 20 kHz then rolling off as shown on figure 4	5.3.1
User equipment voltage spike (spikes may be line-to-line or line-to-ground)	2500 V total including system voltage	5.3.5
Emergency Conditions		
User equipment voltage positive excursion	1060 VDC	5.3.2
User equipment voltage positive excursion duration	2 minutes	5.3.2
User equipment interruption voltage	-100%	5.3.4
User equipment power interruption duration, t_s	2 minutes Default value, see 5.2.2.1	5.3.4
User equipment power interruption duration, t_r	0 ms (no-break supply) 150 ms (not on no-break supply) Default value, see 5.2.2.1	5.3.4

NOTES:

1/ Unless otherwise specified (see 6.2), the individual narrowband components of the voltage ripple shall be limited on a frequency-specific basis as specified on figure 4.

2/ User equipment maximum steady-state voltage ripple is due to the ripple created by the source combined with ripple caused by pulsed loads on the power system.

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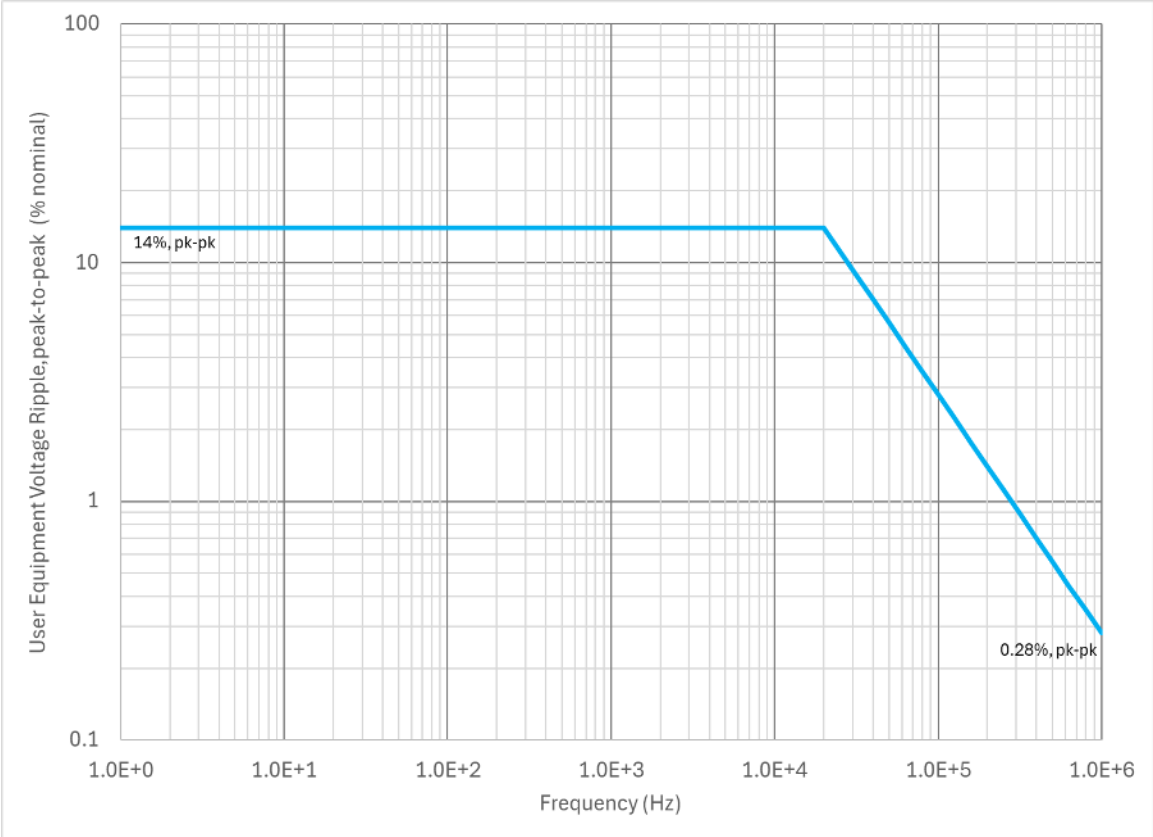


FIGURE 4. User equipment peak-to-peak voltage ripple versus frequency

5.2.1 User equipment susceptibility requirements. User equipment shall operate through the conditions of table II and table III for the type of power specified (see 6.2).

5.2.2 Emergency conditions. User equipment shall withstand and not be damaged by emergency conditions described as power interruptions and excursions. User equipment designated as MCE shall operate through certain emergency conditions as described below.

5.2.2.1 Power interruption. A power interruption experienced by user equipment can occur as a result of normal operations, an equipment casualty, training exercise, or operator error. The length of a power interruption can be specified as t_r (reconfiguration time) or t_s (generator start time), which may be specified (see 6.2). User equipment shall not be damaged due to power interruptions and may require a manual restart. User equipment, not on a no-break source, designated as MCE (see 6.2) shall operate through power interruptions of duration t_r without interruption, not suffer any loss of data, or require a restart or reboot. MCE, not on a no-break power source, after an extended power interruption of t_s, shall be capable of restarting upon restoration of power without operator intervention. t_r will be affected by the electric plant design and configuration. The worst-case maximum duration t_r and t_s values should be determined by the electric plant design agent based upon fault coordination studies and may include bus transfer and recovery time. If t_r is unknown and the equipment is not on a no-break power source, 150 milliseconds will be the t_r power interruption time period specified in table II or table III. If t_s is unknown, 2 minutes will be the t_s power interruption time period specified in table II or table III. T_s and t_r are as specified in MIL-STD-1399-300-1.

5.2.2.2 User equipment voltage positive excursions. User equipment shall be able to tolerate the user equipment voltage positive excursion levels specified in table II or table III without damage. User equipment designated as MCE shall operate without interruption, not suffer any loss of data, or require a restart or reboot through excursions.

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5.2.3 Voltage spikes. User equipment shall remain undamaged, shall not suffer any loss of data, shall not require a restart or reboot, and shall remain fully operational before, during, and after being subjected to voltage spikes with a peak value of 2,500 volts (including system voltage) with a 1.2/50-microsecond wave shape. This means the spike generator shall produce a spike with a rise time of 1.2 microseconds to its peak voltage and a 50-microsecond decay to half the peak voltage when tested open-circuit. Spikes shall be applied line-to-line and line-to-ground. See 5.3.5 for the corresponding verification method.

5.2.4 User equipment constraints. The interface characteristics or the ship's DC electric power system impose certain constraints on the design of shipboard equipment utilizing DC electric power. User equipment constraints and verification methods shall be as specified in 5.2.4 through 5.2.5 and table IV and table V.

TABLE IV. User equipment constraints – type I and III power.

Characteristic	Value: Type I (interface without energy storage)	Value: Type III (interface with regulated energy storage)	Verification Method
User equipment maximum rms current ripple (total rms for frequencies >1 Hz for type I and >120 Hz for type III) (percent of rated current) (Loads between 100 kW and 2 MW)	5% total 1/	5% total	5.3.1
User equipment current ripple frequency content (individual frequencies >1 Hz for type I and 120 Hz for type III)	Per MIL-STD-461 CE101-2 See 5.2.4.1	Per MIL-STD-461 CE101-2 See 5.2.4.1	CE101
User equipment pulsed power magnitude/frequency limits (frequency domain) (1 Hz to 2 kHz) (Loads between 100 kW and 2 MW)	Per figure 5 1/	Per figure 5 1/	5.3.1 and 5.3.6
User equipment power TSD limit (frequency domain) (1 Hz to 2 kHz) (Loads between 100 kW and 2 MW)	5% 1/	N/A	5.3.1 and 5.3.6
User equipment pulsed power deviation limit (time domain) (Loads between 100 kW and 2 MW)	50 kW deviation default unless otherwise approved by NAVSEA 1/ See 5.2.4.3	Not limited provided it does not exceed 100% of specified energy storage power rating. Ramp rate of pulses not to exceed 42.5 MW/s unless otherwise approved by NAVSEA.1/, 3/ See 5.2.4.3	5.3.1 and 5.3.6

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TABLE IV. User equipment constraints – type I and III power – Continued.

Characteristic	Value: Type I (interface without energy storage)	Value: Type III (interface with regulated energy storage)	Verification Method
User equipment surge/inrush current (infrequent, no more than once every 120 seconds)	300% of rated current for user equipment with rated current ≤60 A 180 A for user equipment with 60 A < rated current ≤150 A 120% of rated current for user equipment with rated current >150 A 2/	300% of rated current for user equipment with rated current ≤60 A 180 A for user equipment with 60 A < rated current ≤150 A 120% of rated current for user equipment with rated current >150 A 2/	5.3.1 and 5.3.6
User equipment ramp load (infrequent, no more than once every 120 seconds)	30% of minimum AC source rating in VA on a per-second basis. Maximum step size 70 kW unless otherwise approved by NAVSEA. 3/	42.5 MW/s unless otherwise approved by NAVSEA 3/	5.3.1 and 5.3.6

NOTES:

1/ Values given are for a system fed by a MIL-STD-1399-300-1 or MIL-STD-1399-300-2 interface where power source capacity and behavior of other user equipment is not known. If a type I rectifier is fed from a MIL-STD-1399-300-1 or MIL-STD-1399-300-2 AC system whose power ripple requirements have been relaxed by NAVSEA, then the type I MIL-STD-1399-300-3 DC power and current ripple shall be relaxed to align with the AC values.

2/ If energy storage, including capacitors, is connected directly across the bus, a method to pre-charge the power electronic DC link upon reconnection with energy storage media shall be incorporated within the automated disconnect detailed in the requirement.

3/ Ramp load limits apply to load increases and load decreases.

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TABLE V. Load user constraints – type II power.

Characteristic	Value: Type II (battery-based)	Verification Method
User equipment maximum rms AC current ripple content (total rms for frequencies >120 Hz) (percent of rated DC current) (Loads between 100 kW and 2 MW)	5% total	5.3.1
User equipment current waveform (individual frequencies >120 Hz)	Per MIL-STD-461 CE101-2 See 5.2.4.1	CE101
User equipment pulsed power magnitude/frequency limits (frequency domain) (1 Hz to 2 kHz) (Loads between 100 kW and 2 MW)	Per figure 5	5.3.1 and 5.3.6
User equipment power TSD limit (frequency domain) (1 Hz to 2 kHz) (Loads between 100 kW and 2 MW)	N/A	5.3.1 and 5.3.6
User equipment pulsed power deviation limit (time domain) (Loads between 100 kW and 2 MW)	Not limited provided it does not exceed 100% of specified energy storage power rating. Ramp rate of pulses not to exceed 42.5 MW/s unless otherwise approved by NAVSEA.2/ See 5.2.4.3	5.3.1 and 5.3.6
User equipment surge/inrush current (infrequent, no more than once every 120 seconds)	300% of rated current for user equipment with rated current ≤60 A. 180 A for user equipment with 60 A < rated current ≤150 A. 120% of rated current for user equipment with rated current >150 A. 1/	5.3.1 and 5.3.6
User equipment ramp load (infrequent, no more than once every 120 seconds)	42.5 MW/s unless otherwise approved by NAVSEA 2/	5.3.1 and 5.3.6

NOTES:

1/ If energy storage, including capacitors, is connected directly across the bus, a method to pre-charge the power electronic DC link upon reconnection with energy storage media shall be incorporated within the automated disconnect detailed in the requirement.

2/ Ramp load limits apply to load increases and load decreases.

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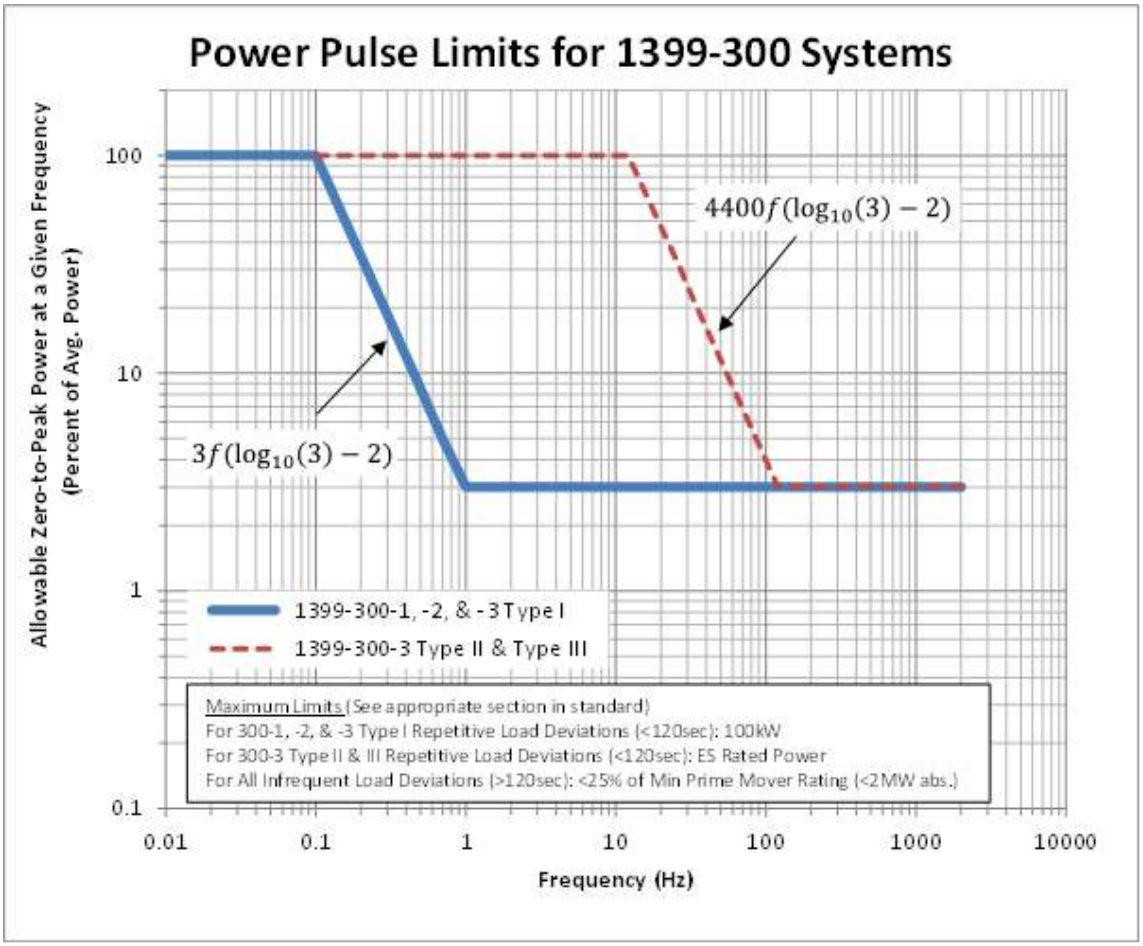


FIGURE 5. User equipment and power source power ripple limits.

5.2.4.1 User equipment current waveform (frequency domain). As specified in table IV and table V, for individual frequencies greater than 1 hertz for type I power and 120 hertz for type II and type III power, differential current limits shall conform to the CE101 limit for surface ships and submarine applications, 60 hertz, specified in MIL-STD-461 with the upper frequency tailored to 20 kilohertz and with the lower frequency tailored to 1 hertz (type I only), where the fundamental current is interpreted as the DC load current. CE101 limits are relaxed based on the DC load current value. The current ripple allowance for type I equipment for frequencies less than 1 hertz shall be derived from the Power Ripple limits shown on figure 5. The current ripple allowance for type II and type III equipment for frequencies less than 120 hertz shall be derived from the Power Ripple limits shown on figure 5.

5.2.4.1.1 Common mode ripple current. The allowed common mode ripple current level shall be 20 decibels less than the differential current limits of MIL-STD-461 CE101 but shall not exceed 3 amperes rms.

5.2.4.2 User equipment maximum rms current ripple. User equipment maximum rms current ripple shall be as specified in table IV or table V from 1 hertz to 20 kilohertz for type I and from 120 hertz to 20 kilohertz for type II and type III and shall be calculated using the same technique as defined for power TSD in 5.2.4.5 making use of the 4-second sliding DFT window for a minimum 100-second sample data set. Hence, rms current for each 4-second sliding window is defined in a similar manner as power TSD whereby current data is processed instead of power data but with frequency content examination up to 20 kilohertz. This requirement applies for loads with a maximum instantaneous real power greater than or equal to 100 kilowatts and less than or equal to 2 megawatts (MW) at the load interface. For loads with a maximum instantaneous real power greater than 2 megawatts, substitute requirements are needed that address the relevant power system's source impedance and response characteristics of the intended platform for all operating modes. The substitute requirements will be as specified (see 6.2) and shall have NAVSEA approval (see 6.3).

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5.2.4.3 User equipment pulsed power deviation limit (time domain). As specified in table IV, for type I power, the user equipment shall limit the instantaneous pulsed power deviation from the average real power level to be less than ±50 kilowatts (kW) over a 1-second integration interval centered on the event or shall be as specified for the MIL-STD-1399-300-1 or MIL-STD-1399-300-2 system feeding the type I rectifier. If a type I rectifier is fed from a MIL-STD-1399-300-1 or MIL-STD-1399-300-2 AC system whose power ripple requirements have been relaxed by NAVSEA, then the type I MIL-STD-1399-300-3 DC power and current ripple shall be relaxed to align with the AC values. If approved by NAVSEA (see 6.2), this limit may be raised to up to 5 percent of the MIL-STD-1399-300-1 or MIL-STD-1399-300-2 AC source power rating if analysis shows this will not negatively affect the source. This requirement applies for loads with a maximum instantaneous real power greater than or equal to 100 kilowatts and less than or equal to 2 megawatts at the load interface. For loads with a maximum instantaneous real power greater than 2 megawatts, substitute requirements are needed that address the relevant power system's source impedance and response characteristics of the intended platform for all operating modes. The substitute requirements will be as specified (see 6.2) and shall have NAVSEA approval (see 6.3). For type II and type III power, the pulsed power deviation is not limited provided it does not exceed 100 percent of specified energy storage power rating or cause non-compliance with the power ripple requirement on figure 5.

5.2.4.4 User equipment pulsed power magnitude/frequency limits (frequency domain). As specified in table IV and table V, between 1 hertz and 2 kilohertz, the user equipment shall limit the magnitude and frequency of the instantaneous pulsed real power by the following constraints:

a. The repetitive pulsed power magnitude/frequency limits are provided on figure 5. The plot presents the acceptable zero-to-peak real power magnitude in watts at a given frequency as a percent of the average real power in watts. The zero-to-peak real power magnitude is determined with the average power component removed from the instantaneous power waveform. The average real power is calculated from the instantaneous real power waveform integrated over a 100-second averaging window. To use figure 5, the instantaneous real power pulsed waveform shall be decomposed into its frequency components after the average real power component has been removed. This is accomplished by using the DFT, incorporating a long window DFT for frequencies below 1 hertz and a short sliding window DFT for frequencies 1 hertz and above. The corresponding recommended procedure for performing the DFT is found in appendix A of MIL-STD-1399-300-1. If the algorithm used to perform the DFT differs from appendix A of MIL-STD-1399-300-1, a listing of the modified algorithm shall be as specified (see 6.2).

b. This requirement applies for loads with a maximum instantaneous real power greater than or equal to 100 kilowatts and less than or equal to 2 megawatts at the load interface. For loads with a maximum instantaneous real power greater than 2 megawatts, substitute requirements are needed that address the relevant power system's source impedance and response characteristics of the intended platform for all operating modes. The substitute requirements will be as specified (see 6.2) and shall have NAVSEA approval (see 6.3).

5.2.4.5 User equipment power TSD limit (frequency domain). For user equipment between 100 kilowatts and 2 megawatts, for type I power, the load shall limit the calculated power TSD to less than 5 percent for frequencies from 1 hertz to 2 kilohertz where the average power shall be calculated over a 100-second window. For type II and type III power, the load shall limit the calculated power TSD to less than 5 percent for frequencies from 120 hertz to 2 kilohertz where the average power shall be calculated over a 100-second window. As specified in appendix A of MIL-STD-1399-300-1, a TSD calculation on each separate 4-second rolling window DFT is performed. The maximum TSD of all 4-second rolling window DFTs is to be the reported TSD value to be compared against the 5 percent requirement.

5.2.4.6 User equipment surge/inrush current. The peak amplitude of the inrush current shall not exceed the limits listed in table IV and table V. Inrush current is characterized as an infrequent event. Infrequent events are defined as events occurring no more than once every 120 seconds.

5.2.4.6.1 DC bus capacitor precharge. If energy storage, including capacitors, is connected directly across the bus, a method to precharge the power electronic DC link upon reconnection with the DC bus shall be incorporated within the user equipment to prevent excessive inrush.

5.2.4.7 User equipment ramp load. As specified in table IV, for type I power, ramp load shall be limited to 30 percent of the minimum source rating on a per second basis. The source rating could result from single or paralleled generators or electronic supplies. If the electric plant architecture allows single generator operation, the source rating is that of the single generator. The resulting average ramp rate can be calculated using consecutive increasing steps. If the ramp load is applied in steps rather than a smooth ramp, the maximum step size shall be

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70 kW. For type II and type III power, per table IV and table V, the ramp load shall be limited to 42.5 megawatts per second (MW/s) unless otherwise approved by NAVSEA (see 6.2 and 6.3). The use of the ramp load allowance is for achieving a desired operational power level and shall only be used for infrequent events. This does not apply for the initial energizing of line filters, converter front end capacitance, transformers, and motors for which the inrush current requirement applies.

5.2.5 Stability (user equipment). User equipment shall operate in a stable manner and shall not cause a resonant or unstable condition to occur at the DC interface or anywhere internal to the user equipment system when supplied with rated voltage.

a. Small signal stability (user equipment). When power source characteristics are specified (see 6.2), the stability of the user equipment shall be assessed against power source characteristics to ensure compatibility with the power source via one of the following: Energy Source Analysis Consortium (ESAC) Criterion with a gain margin of 6 decibels and a phase margin of 45 degrees, Gain Margin/Phase Margin (GMPM) with a gain margin of 6 decibels and a phase margin of 45 degrees, or other method as approved by the Technical Authority. These stability criteria are described in the paper "Comprehensive Review of Stability Criteria for DC Power Distribution Systems," in IEEE Transactions on Industry Applications, Vol. 50, No. 5, pp. 3525-3535, Sept.-Oct. 2014. Small signal analysis shall be performed with and without the impedance of the cables between user equipment and power source when performing stability assessments. The minimum and maximum cable lengths shall be as specified (see 6.2).

b. Input complex impedance (user equipment). The input complex impedance shall be supplied, both minimum and maximum impedance with and without the impedance of the cables between user equipment and power source, as a function of frequency from 0.1 hertz to 100 kilohertz looking into the terminals of the user equipment at the DC bus (graph and text file of Zin versus frequency capturing magnitude and phase). At least 20 data points per decade shall be provided. The minimum and maximum cable lengths shall be as specified (see 6.2).

c. Impedance damping (user equipment). To ensure a well-damped condition, the user equipment impedance characteristics shall be assessed with and without the impedance of the cables between user equipment and power source to ensure a phase slope of no greater than 450 degrees per decade for all expected operating conditions. Both differential and common mode impedance characteristics shall meet these phase slope requirements from 0.1 hertz to 100 kilohertz for differential mode and 0.1 hertz to 1 megahertz for common mode. The minimum and maximum cable lengths shall be as specified (see 6.2).

d. Large signal stability (user equipment). When the characteristics of the power source are known (see 6.2), the large-signal stability of the user equipment shall be assessed against power source characteristics with and without the impedance of the cables to ensure compatibility with the power source. The power source impedance characteristics and minimum and maximum cable lengths shall be as specified (see 6.2). The method of large signal stability analysis shall be approved by NAVSEA (see 6.3). Large signal stability may be assessed by testing in lieu of analysis if approved by NAVSEA.

5.2.6 DC ground isolation (user equipment). Each user equipment input power line at the user interface shall be isolated from hull ground by a minimum DC resistance of 10 megohms.

5.2.7 AC ground isolation (user equipment). The capacitance from line-to-ground at each input power line shall not exceed 0.075 microfarad per kilowatt (μF/kW) of connected load measured at 1 kilohertz. Line-to-ground capacitance from each user equipment terminal to ground shall be balanced to within 10 percent.

5.3 User equipment verification methods. This section provides user equipment verification procedures to verify compliance to the user equipment interface requirements specified in this standard. The hardware and software of an EUT shall be representative of production units. When specified (see 6.2), the verification/test procedure and verification/test report shall be subject to approval by the Technical Authority and the requirements within the verification/test procedure may be tailored as approved by the Technical Authority. Voltage and current measurements shall be in accordance with 4.5. The pulsed load test shall only be performed if specified (see 6.2).

5.3.1 User equipment normal service steady-state voltage range and ripple test. A user equipment's performance shall be verified through testing when provided steady-state power having a steady-state voltage within the range specified in table II or table III for user equipment normal service steady-state voltage range.

a. Apparatus. The following apparatus is recommended for performing this test:

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(1) Power source able to achieve required capacity, specified voltage ripple (unless variable frequency AC source with injection transformer is used), and range of voltage adjustment.

(2) Variable frequency AC source with injection transformer (unless power source can provide specified ripple).

(3) Temperature meter with ±1 °C or better accuracy.

(4) Data acquisition system capable of capturing the voltage waveform, voltage measurement, voltage ripple measurement, current waveform, current measurement, and current ripple measurement. Measurements shall meet the sampling frequency and accuracy requirements of 4.5.

b. Procedure. The EUT shall be operated in all normal operating modes after power has been applied and the EUT has achieved temperature stability for the following test cases:

(1) From powered down condition, apply nominal system voltage ±1 percent, no voltage ripple. Voltage and current inrush waveforms shall be measured and recorded upon energizing the EUT from a powered down condition until the inrush current has subsided.

(2) Apply upper limit of the user equipment normal service steady-state voltage range (or greater), no voltage ripple.

(3) Apply lower limit of the user equipment normal service steady-state voltage range (or less), no voltage ripple.

(4) Apply nominal system voltage ±1 percent, voltage ripple set at the user equipment maximum steady-state voltage ripple.

(5) Apply upper limit of the user equipment normal service steady-state voltage range (or greater), voltage ripple set at the user equipment maximum steady-state voltage ripple.

(6) Apply lower limit of the user equipment normal service steady-state voltage range (or less), voltage ripple set at the user equipment maximum steady-state voltage ripple.

Unless otherwise specified (see 6.2), for test cases (4), (5), and (6) above, the voltage ripple shall consist of a periodic waveform with a fundamental frequency of 1,000±100 hertz.

Once temperature stability has been achieved in each test case, voltage, voltage ripple, current, current ripple, frequency content of the current ripple, and internal equipment temperatures shall be measured and recorded in measurement windows of at least 6 seconds in duration at a maximum of 1-minute intervals for at least 5 minutes in each normal operating mode. The frequency content of the voltage and current ripple shall either be directly measured or calculated from the waveforms and assessed against the user equipment maximum rms AC current ripple content and user equipment current waveform limits. Pulsed power deviation and pulsed power magnitude/frequency limits shall be assessed against 5.2.4.4, 5.2.4.5, and 5.2.4.6 requirements. For each normal operating mode, the EUT shall be verified to function normally.

5.3.2 User equipment voltage transient tolerance test. A user equipment's performance shall be verified through testing when subjected to voltage transients specified in table II or table III.

a. Apparatus. The following apparatus is recommended for performing this test:

(1) Power source(s) able to achieve required capacity, specified voltage ripple, and range of voltage adjustment.

(2) Temperature meter with ±1 °C or better accuracy.

(3) Data acquisition system capable of capturing the voltage waveform, voltage measurement, voltage ripple measurement, current waveform, current measurement, and current ripple measurement. Measurements shall meet the sampling frequency and accuracy requirements of 4.5.

b. Procedure. The EUT shall be energized at a voltage within the user equipment normal service steady-state voltage for a period of not less than 5 minutes then subjected to each of the following test cases:

(1) Increase voltage to the maximum user equipment voltage transient tolerance for a period of the user equipment voltage transient recovery time then return user equipment normal steady-state voltage.

(2) Decrease voltage to the minimum user equipment voltage transient tolerance for a period of the user equipment voltage transient recovery time then return user equipment normal steady-state voltage.

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(3) Increase voltage to the emergency conditions user equipment voltage positive excursion voltage for a period of the user equipment voltage positive excursion time then return user equipment normal steady-state voltage.

For each test case, voltage, voltage ripple, current, current ripple, frequency content of the current ripple, and internal equipment temperatures shall be measured and recorded in measurement windows of at least 6 seconds in duration at a maximum of 1-minute intervals for at least 5 minutes. The EUT shall be verified to not have sustained damage or created a hazardous condition for personnel. If equipment is not MCE, it does not need to operate through the positive excursion voltage test for emergency conditions.

Following completion of the above test cases, nominal system voltage ±1 percent shall be applied. Once the EUT has achieved temperature stability, the EUT shall be operated in all normal operating modes and verified to function normally.

5.3.3 User equipment ground fault test. A user equipment's performance shall be verified through testing when provided steady-state power having a steady-state voltage within the range specified in table II and table III, as applicable, for user equipment normal service steady-state voltage range and with one of the two conductors at ground potential.

a. Apparatus. The following apparatus is recommended for performing this test:

(1) Power source(s) able to achieve required capacity and range of voltage adjustment. The power source shall have a high impedance (greater than 1 megaohm) between its output conductors and ground.

(2) Temperature meter with ±1 °C or better accuracy.

(3) Single pole fused switch for each conductor. The fuse shall be capable of clearing the maximum line-to-line fault current (in case a low impedance already exists within the user equipment between the power system and ground).

(4) Data acquisition system capable of capturing the voltage waveform, voltage measurement, voltage ripple measurement, current waveform, current measurement, and current ripple measurement. Measurements shall meet the sampling frequency and accuracy requirements of 4.5.

b. Procedure. The EUT shall be operated in all normal operating modes with the upper limit of the user equipment normal service steady-state voltage (or greater) for the following test cases:

(1) Positive conductor grounded through the single pole fused switch.

(2) Negative conductor grounded through the single pole fused switch.

For each test case, prior to the start of each test case, the EUT shall be operated without the conductor grounded until the EUT has achieved temperature stability. At the start of the test case, the single pole fused switch shall be closed for the appropriate conductor. After no less than 5 minutes, the ground shall then be removed and the test case repeated for each normal operating mode. The line-to-line voltage, line-to-ground voltage for each conductor, current in each conductor, common-mode current, and internal equipment temperatures shall be measured and recorded at the start of the test and for the duration of the test. During testing, the EUT shall be verified to function normally. The EUT shall fail the test if the fuse associated with one of the fused switches blows.

5.3.4 User equipment interruption tolerance test.

a. Apparatus. The following apparatus is recommended for performing this test:

(1) Power source able to achieve required capability, range of voltage adjustment, and programmability to create power interruptions of specified duration.

(2) Data acquisition system capable of capturing the voltage waveform, voltage measurement, voltage ripple measurement, current waveform, current measurement, and current ripple measurement. Measurements shall meet the sampling frequency and accuracy requirements of 4.5.

(3) Temperature meter with ±1 °C or better accuracy.

b. Procedure. The EUT shall be operated in the following manner for all normal operating modes. Temperatures, duration of power interruption, whether the EUT functions in accordance with 5.2.2.1 and voltage and current waveforms shall be recorded for at least a 500-millisecond period centered on the interruption time period.

(1) Supply the EUT with the nominal system voltage ±1 percent.

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(2) Wait until the EUT has achieved temperature stability.

(3) Interrupt the power for a duration between 1 and 1.1 times the user equipment voltage interruption tolerance time, t_r.

(4) Resume supplying the EUT with the nominal system voltage ±1 percent. Assess the user equipment surge/inrush current that flows upon restoration of voltage against the limits in table IV and table V.

(5) Record user equipment response. If user equipment is MCE, it should operate through the interruption. If user equipment is not MCE, it shall be able to be restarted upon restoration of power.

(6) Restart user equipment if necessary.

(7) Wait until the EUT has achieved temperature stability.

(8) Interrupt the power for a duration between 1 and 1.1 times the user equipment voltage interruption tolerance time, t_s. Assess the user equipment surge/inrush current that flows upon restoration of voltage against the limits in table IV and table V.

(9) Resume supplying the EUT with the nominal system voltage ±1 percent.

(10) Verify that the EUT functions in accordance with 5.2.2.1.

5.3.5 User equipment voltage spike test. The user equipment voltage spike test shall be conducted in accordance with the voltage spike test in the voltage spike (susceptibility) test of MIL-STD-1399-300-1 tailored to a DC system in that the spike does not need to be synchronized with an AC waveform and the spike voltage shall have a peak of 2,500 volts per 5.2.5 including the nominal DC voltage of 850 volts so that the applied spike is 1,650 volts rather than 2,500 volts. The test circuit shall be similar to that for the single-phase, 115-volts rms, type I power voltage spike test circuit configuration of MIL-STD-1399-300-1 except the source shall be a DC source and there is no sync input. The spike shall be applied from line-to-ground and from line-to-line. Note that applying a spike line-to-ground also results in a spike from line-to-line via the coupling capacitors. At a minimum, five spikes shall be applied in each polarity (positive and negative) to each DC line. During testing, the EUT shall be verified to function normally.

Alternatively, the user equipment voltage spike test may be conducted in accordance with the surge immunity test of IEC 61000-4-5 using the test level of 1,650 volts and 1.2/50-microsecond combination wave generator. In either case, during testing, the EUT shall be verified to function normally.

5.3.6 User equipment pulsed load test.

a. Apparatus. The following apparatus is recommended for performing this test.

(1) Power source of required capability and range of voltage adjustment.

(2) Data acquisition system capable of capturing the voltage waveform, voltage measurement, voltage ripple measurement, current waveform, current measurement, and current ripple measurement. Measurements shall meet the sampling frequency and accuracy requirements of 4.5. Test equipment digitizer resolution, measurement skew, and anti-aliasing filters shall be as specified in MIL-STD-1399-300-1, appendix A.

b. Procedure. The user equipment EUT shall be operated in all pulsed modes of operation for the following steady-state voltage test cases:

(1) Nominal system voltage ±1 percent, minimal voltage ripple.

(2) Upper limit of the user equipment normal service steady-state voltage range (or greater).

(3) Lower limit of the user equipment normal service steady-state voltage range (or less).

For each test case, the tester shall verify that the EUT does not exceed the user equipment pulsed power magnitude/frequency limit (frequency domain), user equipment power TSD (frequency domain), user equipment pulsed power deviation limit (time domain), maximum current pulse and user equipment ramp load limits of table IV and table V. The tester shall record voltage and current waveforms. The frequency content of the voltage and current ripple shall either be directly measured or calculated from the waveforms. Test cases shall be repeated for different sets of source limitations (as documented in the test procedures) to ensure robustness in the negotiation process and to verify the EUT does not exceed the negotiated values. User equipment pulsed power magnitude/frequency and user equipment power TSD shall be calculated using the procedure in appendix A of MIL-STD-1399-300-1.

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5.3.7 User equipment line-to-ground resistance and capacitance test.

a. Apparatus. The following apparatus is recommended for performing this test:

(1) Inductance, Capacitance, Resistance (LCR) meter or dedicated capacitance meter.

(2) Megohmmeter insulation tester or insulation testing multimeter capable of reaching 500 VDC.

b. Procedure. The following tests shall be performed with the equipment disconnected from input power and de-energized.

(1) Line-to-ground capacitance test. Line-to-ground capacitance for each power line of the EUT shall be determined by inspection, specification, or measurement in that order and the value shall be recorded and checked against the limit in 5.2.7. Based on available information, the determination may be assisted by one or all of the following topics in their presented order:

(a) Inspect the de-energized EUT input for the presence of filter circuitry and record the line-to-ground capacitance.

(b) Examine the circuit schematic or manufacturer's specification and record the line-to-ground capacitance. A detailed schematic should be included in the test report.

(c) With the EUT de-energized, an approximate determination may be made by temporarily shorting together all input lines and then measuring the capacitance to ground using a capacitance or LCR meter set to 1 kilohertz. Then, divide the measurement value by the number of lines. This will be the line-to-ground capacitance. After the determination, the line-to-line shorts shall be removed.

(2) Line-to-ground resistance test. With the input power disconnected, use a megohmmeter insulation tester (megger) to conduct an insulation resistance test at 500 VDC per the test equipment user manual. With the EUT enclosure grounded, the test shall be conducted on the EUT between each power conductor and ground for a minimum period of 60 seconds after the reading stabilizes for each conductor. The EUT shall tolerate the 500-VDC level without equipment damage, arc-over, or degradation. This test shall be performed with (filter) components connected line-to-ground to determine the voltage withstand capability and insulation resistances. The resistance to ground level as measured with the megohmmeter shall be equal to or greater than 10 megohms (see 5.2.5).

5.4 Power source requirements.

5.4.1 Power source interface. Power source output requirements and verification methods for demonstrating power source operability to the interface characteristics shall be as specified in table VI for type I and type III power and table VII for type II power when supplying an aggregate load with characteristics as specified in table IV and table V up to the total output capacity of the power source. Power source input interface requirements shall be in accordance with the AC power interface standard applicable to the type of power at the input, MIL-STD-1399-300-1 for low voltage AC interfaces and MIL-STD-1399-300-2 for medium voltage AC interfaces.

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TABLE VI. Power source interface requirements – type I and III power.

Characteristic	Value: Type I (interface without energy storage)	Value: Type III (interface with regulated energy storage)	Verification Method
Power source nominal voltage	850 VDC	850 VDC	N/A
Power source maximum steady-state voltage ripple (peak-to-peak percent of nominal system voltage)[1/, 2/]	10.0% up to 20 kHz then rolling off as shown on figure 6	10.0% up to 20 kHz then rolling off as shown on figure 6	5.5.1
Power source normal service voltage range: 0 to rated power	782 to 918 VDC (-8 to +8%)	799 to 884 VDC (-6 to +4%)	5.5.1
Power source voltage transient tolerance	646 to 1045 VDC (-24 to +23%)	731 to 926 VDC (-14 to +9%)	5.5.4
Power source voltage transient recovery time	2 seconds	0.25 second	5.5.4

NOTES:
1/ Unless otherwise specified (see 6.2), the individual narrowband components of the voltage ripple shall be limited on a frequency-specific basis as specified on figure 6.
2/ Power source maximum steady-state voltage ripple is due to the ripple created by the source combined with ripple caused by pulsed loads on the power system.

TABLE VII. Power source interface requirements – type II power.

Characteristic	Value: Type II (battery-based)	Verification Method
Power source nominal voltage	875 VDC	N/A
Power source maximum steady-state voltage ripple (peak-to-peak percent of nominal system voltage)[1/, 2/]	10% up to 20 kHz then rolling off as shown on figure 6 [3/, 4/]	5.5.1
Power source normal service voltage range, charged to discharged [5/]	680 to 1000 VDC (-20 to +18%)	N/A
Power source voltage transient tolerance	650 to 1060 VDC (-24 to +24%)	5.5.4
Power source voltage transient recovery time	2 seconds at 650 VDC 30 seconds at 1060 VDC	5.5.24

NOTES:
1/ Unless otherwise specified (see 6.2), the individual narrowband components of the voltage ripple shall be limited on a frequency-specific basis as specified on figure 6.
2/ Power source maximum steady-state voltage ripple is due to the ripple created by the source combined with ripple caused by pulsed loads on the power system.
3/ 0 percent if source is feeding a steady load and voltage is entirely determined by battery voltage.
4/ During maintenance charging, balancing, or float charging (if applicable), ripple voltage is maximum of 0.5 percent (rms) of the DC voltage; ripple current does not exceed 5 amperes per 100 ampere-hour of rated capacity.
5/ Includes state of charge and effects of supplying user equipment.

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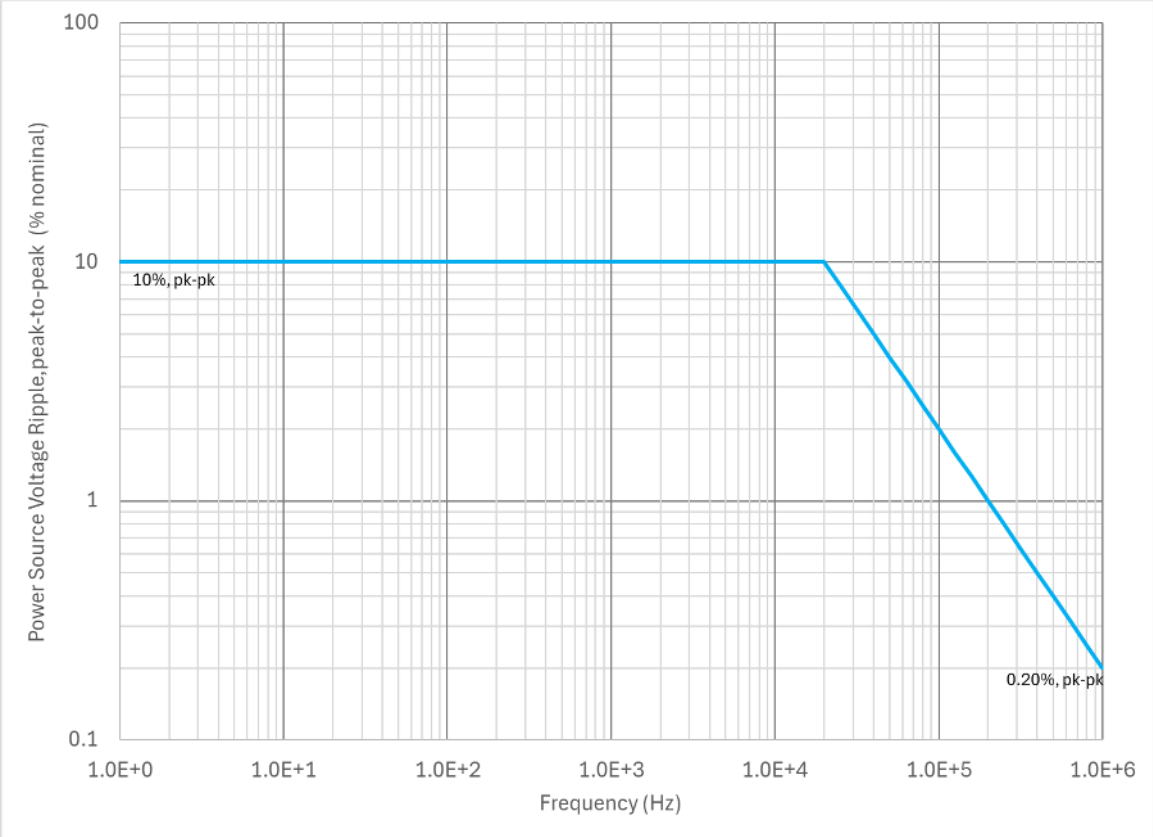


FIGURE 6. Power source peak-to-peak voltage ripple versus frequency.

5.4.2 Power source maximum rms current ripple. Power sources shall meet CE101 for conducted emission of power leads, 30 hertz to 10 kilohertz, on the DC output when tested against a resistive load. The DC output differential current limit is CE101 limit for surface ships and submarine applications, 60 hertz, from MIL-STD-461 with the upper frequency tailored to 20 kilohertz and with the lower frequency tailored to 1 hertz (type I only), where the fundamental current is interpreted as the DC load current. Both differential and common mode currents shall be measured on the DC outputs. The allowed common mode current level shall be 20 decibels less than the differential current limits of MIL-STD-461 CE101-2 but shall not exceed 3 amperes rms.

5.4.3 Voltage spike. Power sources shall remain undamaged, shall not suffer any loss of data, shall not require a restart or reboot, and shall remain fully operational before, during, and after being subjected to voltage spikes on their output with a peak value of 2,500 volts (including system voltage) with a 1.2/50-microsecond wave shape. This means the spike generator shall produce a spike with a rise time of 1.2 microseconds to its peak voltage and a 50-microsecond decay to half the peak voltage when tested open circuit. Spikes shall be applied line-to-line or line-to-ground. See 5.5.3 for the corresponding verification method.

5.4.4 Stability (power source). Power sources shall operate in a stable manner and shall not cause a resonant or unstable condition to occur at the DC interface or anywhere internal to the power source when supplying rated voltage to the connected load from zero load to the full rated load of the power source.

a. Small signal stability (power source). When user equipment characteristics are specified (see 6.2), the stability of the power source shall be assessed against user equipment characteristics to ensure compatibility with the user equipment via one of the following: ESAC Criterion with a gain margin of 6 decibels and a phase margin of 45 degrees, GMPM with a gain margin of 6 decibels and a phase margin of 45 degrees, or other method as approved by NAVSEA. These stability criteria are described in the paper “Comprehensive Review of Stability Criteria for DC Power Distribution Systems,” in IEEE Transactions on Industry Applications, vol. 50, no. 5, pp. 3525-3535, Sept.-Oct. 2014. Small signal analysis shall be performed with and without the impedance of the cables between

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user equipment and power source when performing stability assessments. The minimum and maximum cable lengths shall be specified (see 6.2).

b. Output complex impedance (power source). The output complex impedance shall be supplied, both minimum and maximum impedance with and without the impedance of the cables between user equipment and power source, as a function of frequency from 0.1 hertz to 100 kilohertz looking into the terminals of the power source at the DC bus (graph and text file of Zin versus frequency capturing magnitude and phase). At least 20 data points per decade shall be provided. The minimum and maximum cable lengths shall be specified (see 6.2).

c. Impedance damping (power source). To ensure a well-damped condition, the power source impedance characteristics shall be assessed with and without the impedance of the cables between user equipment and power source to ensure a phase slope of no greater than 450 degrees per decade for all expected operating conditions. Both differential and common mode impedance characteristics shall meet these phase slope requirements from 0.1 hertz to 100 kilohertz for differential mode and 0.1 hertz to 1 megahertz for common mode. The minimum and maximum cable lengths shall be specified (see 6.2).

d. Large signal stability (power source). The large-signal stability of the power source shall be assessed against user equipment characteristics with and without the impedance of the cables to ensure compatibility with the user equipment. The user equipment characteristics, which may include, but are not limited to, constant power load, constant current load, or a resistive load, along with associated input filters and the minimum and maximum cable lengths shall be specified (see 6.2). The method of large signal stability analysis shall be approved by NAVSEA (see 6.2). Large signal stability may be assessed by testing in lieu of analysis if approved by NAVSEA.

5.4.5 Parallel operations. Power sources shall function normally and meet interface requirements specified herein when paralleled with other power sources employing load sharing or other power management method(s) as specified (see 6.2). Verification requirements shall be as specified (see 6.2).

5.4.6 DC ground isolation (power source). Each power source DC output power line at the power source interface shall be isolated from hull ground by a minimum DC resistance of 10 megohms.

5.4.7 AC ground isolation (power source). The leakage capacitance from each line-to-ground at each DC output power line shall not exceed 0.02 microfarad per kilowatt measured at 1 kilohertz. Line-to-ground capacitance from each power source output terminal to ground shall be balanced to within 10 percent.

5.5 Power source verification methods. This section provides power source verification procedures to verify compliance to the power source interface requirements specified in this standard. For power sources verified through testing, the hardware and software of an EUT shall be representative of production units. Verification/test procedure and verification/test report requirements may be tailored as specified (see 6.2) as approved by the Technical Authority. Voltage and current measurements shall be in accordance with 4.5.

5.5.1 Power source voltage regulation test.

a. Apparatus. The following apparatus is recommended for performing this test:

(1) Programmable load capable of operation over the power source steady-state voltage range: 0 to rated power, with adjustable load up to the power source rated power, and the ability to inject user equipment maximum rms ripple current.

(2) Temperature meter with ±1 °C or better accuracy.

(3) Data acquisition system capable of capturing the voltage waveform, voltage measurement, voltage ripple measurement, current waveform, current measurement, and current ripple measurement. Measurements shall meet the sampling frequency and accuracy requirements of 4.5.

b. Procedure. The EUT shall be operated with the following commanded reference voltage and load test cases:

(1) Nominal system voltage and 8 to 12 percent load.

(2) Lower limit of power source steady-state voltage range: 0 to rated power.

(3) Upper limit of power source steady-state voltage range: 0 to rated power.

(4) Nominal system voltage and 95 to 100 percent load.

(5) Nominal system voltage and 8 to 12 percent load with user equipment maximum rms current ripple.

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(6) Lower limit of power source steady-state voltage range: 0 to rated power with user equipment maximum rms current ripple.

(7) Upper limit of power source steady-state voltage range: 0 to rated power with user equipment maximum rms current ripple.

(8) Nominal system voltage and 95 to 100 percent load with user equipment maximum rms current ripple.

(9) Lower limit of user equipment steady-state voltage range: 0 to rated power and 95 to 100 percent load with user equipment maximum rms current ripple.

(10) Upper limit of user equipment steady-state voltage range: 0 to rated power and 95 to 100 percent load with user equipment maximum rms current ripple.

Unless otherwise specified (see 6.2), for test cases (5) through (10), the current ripple shall consist of a periodic waveform with a fundamental frequency of 1,000±100 hertz.

For each test case, once the EUT has achieved temperature stability, voltage, voltage ripple, current, current ripple, and internal equipment temperatures shall be measured and recorded in measurement windows of at least 6 seconds in duration at a maximum of 1-minute intervals for no less than 5 minutes. The frequency content of the voltage and current ripple shall either be directly measured or calculated from the waveforms. Once the EUT has achieved temperature stability, the EUT shall be verified to function normally.

5.5.2 Power source ground fault test.

a. Apparatus. The following apparatus is recommended for performing this test:

(1) Programmable load capable of operation over the power source steady-state voltage range: 0 to rated power, with adjustable load up to the power source rated power.

(2) Temperature meter with ±1 °C or better accuracy.

(3) Single pole fused switch for each conductor. The fuse shall be capable of clearing the maximum line-to-line fault current (in case a low impedance already exists within the power source between the power system and ground).

(4) Data acquisition system capable of capturing the voltage waveform, voltage measurement, voltage ripple measurement, current waveform, current measurement, and current ripple measurement. Measurements shall meet the sampling frequency and accuracy requirements of 4.5.

b. Procedure. The EUT shall be operated with the following commanded reference voltage and grounding test cases:

(1) Upper limit of power source steady-state voltage range with positive conductor grounded through the single pole fused switch.

(2) Upper limit of power source steady-state voltage range with negative conductor grounded through the single pole fused switch.

For each test case, prior to the start of each test case, the EUT shall be operated without the conductor grounded until the EUT has achieved temperature stability. At the start of the test case, when the single pole-fused switch is closed for the appropriate conductor, the line-to-line voltage, line-to-ground voltage for each conductor, current in each conductor, common mode current, and internal equipment temperatures shall be measured and recorded for the duration of the test. After no less than 5 minutes, the ground shall then be removed and the test case repeated for each normal operating mode. During testing, the EUT shall be verified to function normally. The EUT shall fail the test if the fuse associated with one of the fused switches blows.

5.5.3 Source voltage spike test. The power source voltage spike test shall be conducted in accordance with the voltage spike test in the voltage spike (susceptibility) test of MIL-STD-1399-300-1 tailored to a DC system in that the spike does not need to be synchronized with an AC waveform and the spike voltage shall have a peak of 2,500 volts per 5.4.3 including the nominal DC voltage of 850 volts so that the applied spike is 1,650 volts rather than 2,500 volts. The test circuit shall be similar to that of the single-phase, 115-volts rms, type I power voltage spike test circuit configuration of MIL-STD-1399-300-1, except the source shall be a DC source and there is no sync input. The spike shall be applied from line-to-ground and from line-to-line. Note that applying a spike line-to-ground also results in a spike from line-to-line via the coupling capacitors. At a minimum, five spikes shall

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be applied in each polarity (positive and negative) to each DC line. During testing, the EUT shall be verified to function normally.

Alternatively, the user equipment voltage spike test may be conducted in accordance with the surge immunity test of IEC 61000-4-5 using the test level of 1,650 volts and 1.2/50-microsecond combination wave generator. In either case, during testing, the EUT shall be verified to function normally.

5.5.4 Power source pulse test.

a. Apparatus. The following apparatus is recommended for performing this test:

(1) Controllable pulsed load capable of generated pulses with a pulse width between 100 milliseconds and 1 second.

(2) Programmable load capable of operation over the power source steady-state voltage range: 0 to rated power, with adjustable load up to the power source rated power.

(3) Data acquisition system capable of capturing the voltage waveform, voltage measurement, voltage ripple measurement, current waveform, current measurement, and current ripple measurement. Measurements shall meet the sampling frequency and accuracy requirements of 4.5.

b. Procedure. The base load measured in kilowatts shall be set equal to 90 percent of the difference between rated load of the power source measured in kilowatts minus the appropriate power source maximum current pulse measured in kiloamperes multiplied by the nominal system voltage. For type III systems, the power source consists of the combination of energy storage systems and their intervening power electronics and the connecting AC/DC rectifiers. The power source EUT shall be operated in the following test cases:

(1) Nominal system voltage ±1 percent, programmable load set to the base load, and pulse between 95 and 100 percent of the user equipment pulsed power deviation limit where the user equipment power rating matches that of the power source.

(2) Upper limit of the power source normal service steady-state voltage range, programmable load set to the base load, and pulse between 95 and 100 percent of the user equipment pulsed power deviation limit where the user equipment power rating matches that of the power source.

(3) Lower limit of the power source normal service steady-state voltage range, programmable load set to the base load, and pulse between 95 and 100 percent of the user equipment pulsed power deviation limit where the user equipment power rating matches that of the power source.

The EUT shall maintain power quality before, during, and following the pulse. Voltage and current waveforms shall be recorded. The frequency content of the voltage and current ripple shall either be directly measured or calculated from the waveforms. The procedure shall be repeated for each operational mode of the EUT.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. This standard is intended to be used in designing surface ship low voltage DC electrical power system sources and user equipment.

The voltage spike and electromagnetic pulse (EMP) protection requirements enhance the ability of power sources and user equipment to remain undamaged following exposure to EMP. While these requirements reduce the susceptibility of equipment to EMP, they must be incorporated into an overall systems engineering approach to survivability in order for a ship to be capable of continued operation following exposure to EMP. Refer to equipment specifications for requirements.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Verification methods (see 4.1.1 and 4.1.2).

c. Nominal system voltage if other than specified (see 4.2).

d. Normal operating modes (see 4.3).

e. Alternate measurement methods if other than specified (see 4.5).

f. Sampling rate for common mode currents and voltages (see 4.5.1).

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g. Requirements for deviation and tailoring requests (see 4.6).

h. Conformance tests (see 4.7).

i. Characteristics of AC power system feeding type I system (see 5.1.2).

j. When individual narrowband components of the voltage ripple are limited on a frequency-specific basis differently than as specified on figure 4 (see table II and table III).

k. Type of power specified for user equipment (see 5.2.1).

l. When an alternate value for the user equipment voltage interruption tolerance time may be used in lieu of the value specified in table II or table III (see 5.2.2.1).

m. If equipment is designated as MCE (see 5.2.2.1).

n. Substitute requirements to address the relevant power system's source impedance and response characteristics of the intended platform for all operating modes for loads with a maximum instantaneous real power greater than 2 megawatts (see 5.2.4.4 and 5.2.4.5.b).

o. If the algorithm used to perform DFT differs from MIL-STD-1399-300-1 (see 5.2.4.5.a).

p. When the ramp load is not limited to 42.5 MW/s for type II and type III power (see 5.2.4.7).

q. Data associated with stability requirements such as power source and/or user equipment impedance requirements, minimum and maximum cable lengths, whether or not large signal stability analysis is required, and method of large signal stability analysis, if required (see 5.2.5 and 5.4.5).

r. Requirements for Technical Authority approval of verification/test procedures and verification/test reports derived from tests specified (see 5.3 and 5.5).

s. Requirement to perform the pulsed load test (see 5.3).

t. Voltage ripple or current waveform and frequency if other than specified (see 5.3.1.b and 5.5.1).

u. When individual narrowband components of the voltage ripple are limited on a frequency-specific basis differently than as specified on figure 6 (see table VI and table VII).

v. Power source load sharing or power management method(s) and parallel operation verification requirements (see 5.4.5).

6.3 Deviation, waiver, and tailored requirement requests. Requests for deviations, waivers, and tailored requirements are to be submitted for approval to the NAVSEA Technical Authority for Ship Electrical Power Systems (NAVSEA 05Z3).

6.4 Subject term (key word) listing.

Electrical load
Electrical source
Electrical transients
Inrush current
LVDC
Power quality
Pulsed load
Ripple
Voltage spike

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH

Preparing activity:
Navy – SH
(Project 1905-2021-002)

Review activities:
Army – MI, MT

Civil agency:
GSA – FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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